BPV: Measuring Fantasy Basketball Value

The method, the website, and every test behind the numbers

What this report covers
1. The final BPV formula and a verdict on every question we tested
2. The website: Watchlist, Trade Analyzer and Waiver Wires, and how each number is computed
3. The ideas: availability, replacement level, types of missed games, recent form, new roles, rookies, injury risk
4. How we tested: data, backtest design, error measures, resampling, and no-peeking model comparisons
5. The ideal percentages: recent-form weight, New Role haircut and weight, availability settings, early-season blending
6. Which method was best, how sure we are, what's still open, and what to try next

Data: NBA regular seasons 2019-20 through 2025-26 (every player-game and team-game), plus all NBA drafts. Generated September 28, 2026.

1. Summary

BPV (Basketball Performance Value) estimates how many fantasy points per team game you get from having a player on your roster in an ESPN points league. It is built on one idea: **fantasy value is not the same as basketball value.** A fantasy manager gets a player's production when he plays, and can often stream a free agent when he doesn't. The final formula is:

$$BPV = p \times F + (1 - p) \times q \times R$$

Symbol	Meaning	How it's computed
F	Production: fantasy points per game played	Season average + 0.3 × recent minutes change + 0.1 × recent per-minute change (last 5 games). New Role Value for players whose minutes jumped. Early in the season, blended (8 games' worth) with the saved preseason projection, or last season, or a draft-slot baseline (10 games). Between seasons: last season + age and roster adjustments, blended 37.5% with ESPN's preseason projection. In season: recalibrated.
p	Availability: share of team games he's expected to play	Recent seasons, each older season counting half as much, blended with 123 games of league-average availability (about 74%). Before a season: ESPN projected games × 0.86 / 82 where available. In season: adjusted for injuries on the official report, by injury type.
q	Share of missed games that come in long, IR-able stretches (5+ straight games)	League average, about 72%
R	Replacement level: what a free agent gives you	Fantasy points per game of the 130th-best player (10 teams × 13 roster spots); about 27.5 in 2025-26

Verdicts

Every question below was tested on real data. "Established" means the result held up on a season the method never saw and under resampling. "Not established" is also a firm answer: the data cannot distinguish the options.

Question	Best method found	Verdict
Should recent games count extra?	Yes: 10–20% weight on the last 5 games played	**Established** (cuts error about 3%)
Is 20% the ideal weight?	Anything from 10% to 20% is equally good; 15% fit best most often	**Not established**: 20% and 15% are a statistical tie
Last 5 games played, or the team's last 5?	Last 5 games played	Slightly better; small difference
Do recent minutes changes last more than hot/cold shooting?	Yes: about 30% of a minutes change lasts vs about 10% of a shooting change	**Established**
Does splitting minutes from shooting predict better than plain 80/20?	Split is at least as good	**Not established** (gain too small to confirm)
Does New Role Value beat 80/20 after a minutes jump?	Yes, by about 1 FPTS per game	**Established**
Is the 90% haircut right?	0.90 with 60% weight fit best, but 0.80–1.10 all fit about equally	**Not established**
Best availability estimate?	Recent seasons weighted, shrunk heavily to the league average	**Established**, but only about 1 point of games better than the average
Is career GP% useful?	No better than assuming every player is average	**Established** (no significant difference)
Should missed games count as zero?	No: count streamed free-agent points for long absences	Logical argument; supports how managers actually play

Question	Best method found	Verdict
Are erratic vs long absences a player trait?	Can't be measured reliably from box scores	**Not determinable** with this data
How to value players early in the season?	Blend with last season (8 games) or a draft-slot baseline for rookies (10 games)	**Established** (large error reduction)
Does age matter between seasons?	Yes: young players gain up to +2.1 FPTS/game, 33+ lose about 3.7	**Established** (4.7% less error, all 4 seasons)
Do older seasons help a projection?	No: last season alone is best (tested two ways)	**Established** (no gain)
Does BPV beat a published baseline?	Last season + age beat Basketball-Reference's projection system	**Established** for preseason production
Exponential decay instead of last 5 games?	Ties the current method	No gain; recent form is at its ceiling
Which preseason model ranks players best?	The BPV + ESPN hybrid (0.686), ahead of ESPN (0.671) and plain BPV (0.664)	**Established** vs ESPN (+0.016, interval +0.007 to +0.024)
Does BPV beat ESPN?	Per game, yes; on season-total rankings and drafts, plain BPV lost to ESPN	**Honest answer: only the hybrid beats ESPN**
Which preseason model drafts best?	A BPV version won all 4 seasons of simulated drafts (2 with age, 2 without)	**Suggestive, not conclusive**
Does a Cox injury-risk model improve availability?	Built from box-score absences and tested: it doesn't beat the current estimate	**No** (not added)
Does next season's roster matter?	Yes: more minutes of competition on the new team lowers production	**Established** (added)
Does a teammate's long-term injury raise a player's value?	Yes, when the injury is known (+0.034 FPTS/g per lost minute)	**Established** for known injuries only
Are the 80% likely ranges trustworthy?	In season 79.8% coverage; between seasons 72–82%	**Calibrated** (added)
Do official injury reports help?	Yes: a serious injury listing predicts a long absence with no hindsight	**Established** (automatic in season)
Are BPV's in-season projections biased?	Slightly too extreme (young players improve in season); recalibration fixes it	**Established** (added)
Do bad teams rest regulars in the fantasy playoffs?	Yes: veterans on bad teams play 44 points less of their games	**Established** (added to expected playoff FPTS)

2. The website

The site runs locally (**python app.py**, then open http://127.0.0.1:8080). It downloads whole seasons of NBA game logs from stats.nba.com, one request per season, and caches them on disk. Completed seasons are kept permanently, and the current season refreshes every 6 hours. All BPV math runs on the server, so every page shows the same numbers.

Season modes

- **In-season:** a game was played in the last 14 days. Recent form, role-change detection and early-season blending are on.
- **Offseason:** the season has ended (today's state: 2025-26 is over). Values use the full season with recent form turned off, because a team's last 5 games of a season are distorted by resting and tanking, plus age and roster-crowding adjustments (Section 14). These values project to next season.
- **Preseason:** a new season has no games yet. The previous season is used until games are played.

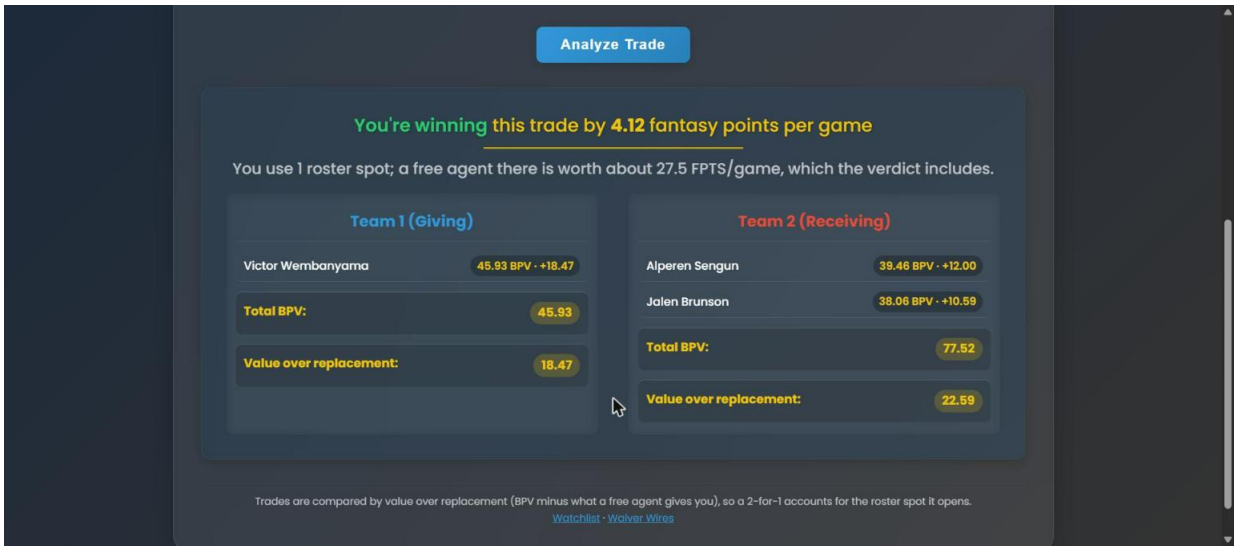
Watchlist (home page)



The Watchlist: BPV, Heat Check Index, the full breakdown, likely range and fantasy-playoff games. Between seasons the note shows the age and roster adjustments.

- **BPV** and **rank** among all players.
- **Heat Check Index (HCI):** BPV as a percentage of the league's best BPV (the best player = 100).
- **Production, season average and last-5 average**, with a note on how production was computed (season average, recent form, new role detected, blended with last season, or draft-slot baseline).
- **Expected availability**, games missed this season, and **value over a free agent** (BPV − R).
- The full formula with the player's own numbers filled in, and his per-game box-score averages.
- A **top-25 table**; clicking a row opens that player.
- Newly drafted players with no NBA games show a projection from their draft range instead of 0.
- **Likely range (80%)** for production, and **fantasy-playoff games** (his team's games in the last 4 weeks; 13–16 in 2026-27).
- Between seasons, players are shown on **next season's team** (from current NBA rosters), and unsigned players as FA.
- League size and roster size are settings (LEAGUE_TEAMS, LEAGUE_ROSTER_SIZE; default 10 and 13), which set the replacement level.
- In season: the **official injury report** status and reason, with the chance of playing (Probable 91%, Questionable 49%, Doubtful 2%), **this week's expected FPTS**, and **expected fantasy-playoff FPTS**, which allows for late-season resting on bad teams.
- Between seasons: **ESPN's projection** (per game and games) next to BPV's, blended into BPV as tested.

Trade Analyzer

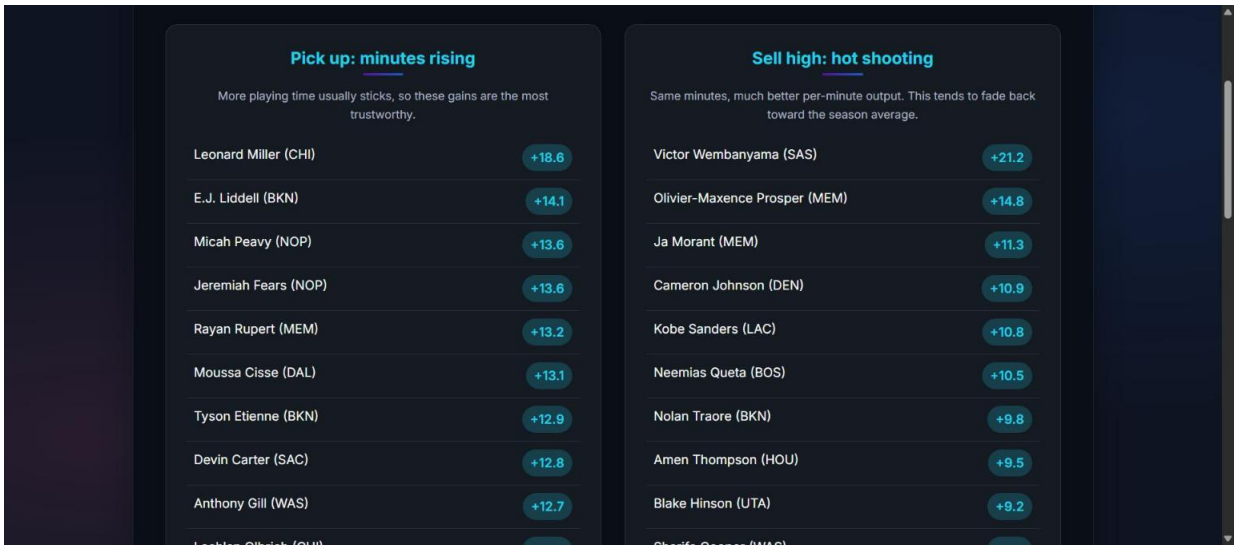


A 2-for-1: the receiving side has 31.6 more raw BPV, but after accounting for the roster spot the edge is 4.1 FPTS per game.

Add players to each side and press Analyze. The verdict compares **value over replacement** (BPV − R) instead of raw BPV. Why: in a 2-for-1, the side that gets fewer players frees a roster spot, and that spot is filled by a free agent worth R. Summing raw BPV ignores this and always favors the side getting more players. Clicking a player removes him (this also fixed a bug where names with apostrophes, like De'Aaron Fox, couldn't be removed).

Caution: BPV's typical error predicting a single player's rest-of-season average is about 4 FPTS per game. A trade edge of 1–3 points per game is within that noise. Treat small edges as "about even."

Waiver Wires



Waiver Wires, split by why a player's last 5 games changed.

Each player's last-5 change is split into a **minutes part** (per-minute production × change in minutes) and a **per-minute part** (everything else: shooting and luck). Four lists follow from the finding in Section 11 that minutes changes last and shooting swings fade:

List	Rule	Why
Pick up: minutes rising	Minutes part ≥ threshold	More playing time usually sticks

List	Rule	Why
Sell high: hot shooting	Per-minute part ≥ threshold, minutes steady	Hot shooting in the same minutes tends to fade
Buy low: cold shooting	Per-minute part ≤ −threshold, minutes steady	Cold shooting in the same minutes tends to recover
Drop candidates: minutes falling	Minutes part ≤ −threshold	Lost playing time usually lasts; check the news first

The threshold (default 4 FPTS per game) is adjustable. The site can't see your league's rosters, so the lists cover the whole league.

Manual inputs

data/overrides.json holds what box scores can't know. Injuries use **{"out_for_season": true}** or **{"out_games": 10}**; those games are treated as one long, IR-able absence, so they earn the free-agent credit. Role changes use **{"since": "2026-01-15"}**, which applies New Role Value from that date. This replaces the first version's hard-coded list of 40 players forced to BPV 0, which described 2024-25 injuries. A known injury also helps his teammates: in season they get +0.034 FPTS per game for each minute per game he played; between seasons his minutes are removed from his team's roster crowding (Section 14).

3. The core idea

Imagine a player who is great at his job and gets the crowd hyped, but misses a lot of games. Fans love him, but a fantasy manager doesn't get production worth the hype. Now imagine a slightly worse player who stays quiet but consistently produces and shows up to nearly every game. The fans wouldn't go crazy over him, but a fantasy manager would. The goal isn't to find the best basketball player; it's to find the player who gives a manager the most expected value.

Terms

Term	Meaning
ADP	Average draft position: a player's average pick number across many drafts. It shows where players are taken, not where they should be.
FPTS	Fantasy points, using ESPN's default points scoring (Section 4).
Roster	ESPN standard: 13 players, of which 10 start (PG, SG, SF, PF, C, G, F and 3 UTIL) and 3 sit on the bench.
IR	Injured reserve: a slot outside the 13. Only players with the O (Out), IR or IL tag can be placed there, so Questionable and Day-to-Day players can't. The number of IR slots varies by league.
Free agent / waivers	Undrafted or dropped players anyone can add. A recently dropped player goes through waivers; ESPN orders claims either by resetting weekly to reverse standings or by moving a team to the back after each claim.
Replacement level (R)	What a typical free agent produces. With 10 teams × 13 spots, 130 players are rostered; the 130th-best averages about 25–27.5 FPTS per game.

4. The scoring formula

$$\text{FPTS} = \text{PTS} + \text{REB} + 2{\cdot}\text{AST} + 4{\cdot}\text{STL} + 4{\cdot}\text{BLK} - 2{\cdot}\text{TO} + 2{\cdot}\text{FGM} - \text{FGA} + \text{FTM} - \text{FTA} + \text{3PM}$$

This is ESPN's default points scoring. It values points, rebounds and assists, and it also rewards efficiency: a made two is worth +3, a made three +5, a made free throw +1, and every miss −1. BPV never changes these weights. The point is not "what the scoring should be" but "given the real scoring, what is a player worth." BPV always works in **FPTS per game**; season totals already include games missed, so multiplying a total by availability

would count absences twice.

5. Availability and replacement level

Player A averages 45 FPTS and plays 60% of games: 45 × 0.60 = 27. Player B averages 30 and plays 90%: 30 × 0.90 = 27. They look identical, but a missed game isn't worth zero. When A is out, you can put him on IR and pick up a free agent. If a free agent gives you 20 FPTS:

- Player A: 45 × 0.60 + 20 × 0.40 = **35**
- Player B: 30 × 0.90 + 20 × 0.10 = **29**

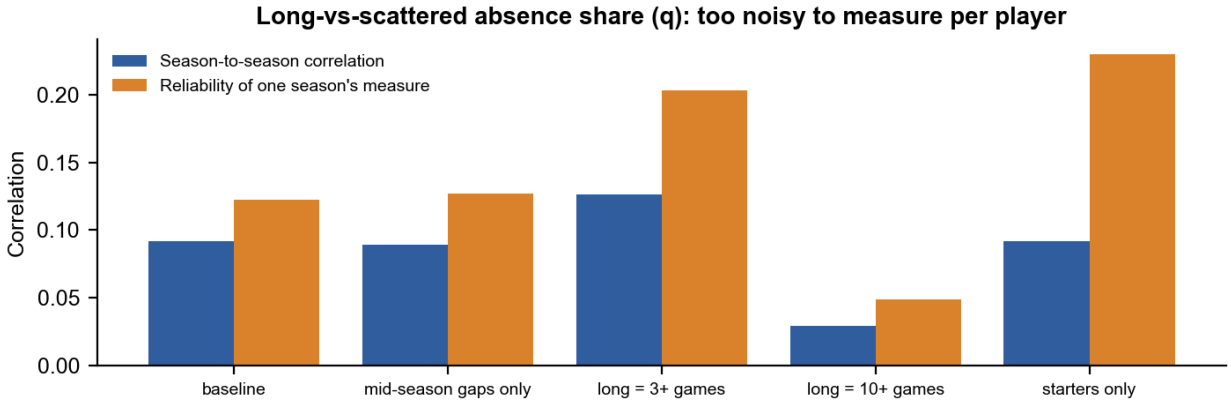
$$\text{Value} = p \times F + (1 - p) \times R = R + p \times (F - R)$$

Multiplying production by availability is the special case R = 0, which assumes you get nothing when a player sits. The second form shows what really drives value: how far a player is above a free agent, times how often he plays. Two limits apply: leagues have only one or two IR slots (if one is already used, the next injured player's missed games really are worth zero), and benching a player doesn't open a roster spot.

6. Types of missed games

Player D misses 30 straight games with one injury; Player E misses 30 scattered games. Their availability is identical, but D goes on IR once and you stream free agents in his place, while E cycles between Questionable and Out, can't be placed on IR while Questionable, and forces awkward lineup decisions. So only long absences get the free-agent credit, through q: the share of missed games that come in stretches of 5 or more straight team games.

How we measured it. For every player-season we walked his games against his team's schedule and recorded every run of consecutive team games he missed. Around a trade, the gap is measured on whichever team's schedule gives the shorter gap; in a debut season, games before his first appearance aren't counted. Spot checks: Giannis Antetokounmpo 2025-26 played 36 and missed 46 (82 possible); Nikola Jokić played 65 and missed 17; traded players came out at 80–83 possible games.



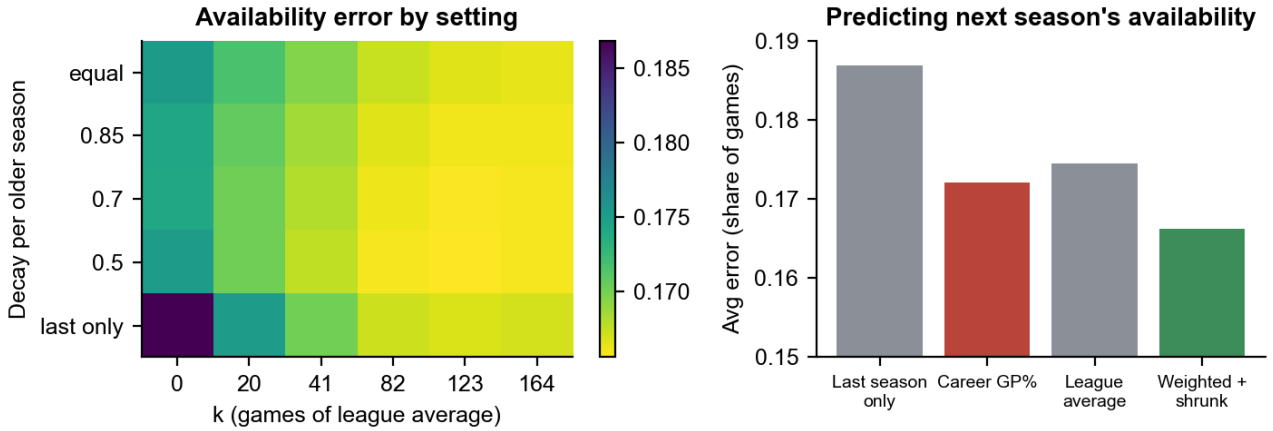
Blue: does a player's q repeat the next season? Orange: does q even agree between the two halves of one season?

Across rotation players, about **72%** of missed games come in long stretches. But a player's own q barely repeats from one season to the next (correlation 0.03–0.13 depending on the definition). That alone would suggest "not a trait," but the orange bars show the measure is too noisy to judge: the two halves of the same season barely agree either, which puts the reliability of one season's q at about 0.05–0.23. Box scores can't separate injuries from rest days, suspensions or coach's decisions, which adds noise.

Verdict: Whether a player tends to miss games erratically is **not determinable** from box-score data. BPV therefore uses the league-average q (72%) for everyone, which is the right choice for an unreliable measure. Known long injuries can be entered manually. Injury-report data would be needed to test the Player D vs E idea properly.

7. Estimating availability

We tested how best to predict a regular player's (20+ minutes per game) share of games played next season from his history. Settings tried: how much each older season counts (all equal; 0.85, 0.7 or 0.5 per year back; or last season only) and how many games of league-average availability to blend in (k = 0 to 246).



Left: error for each setting (lighter is better). Right: methods compared on seasons they weren't tuned on.

Method	Avg error (share of games)	Result
Last season only	0.187	Worst
Career GP% (first version of BPV)	0.172	Not significantly different from the league average
Everyone gets the league average	0.174	Baseline
Recent seasons weighted + shrunk to the league average	0.166	Best; better than the average by 0.004–0.013 (95% interval)

The error surface is flat near the best: decay 0.5–0.7 per season with k = 123–164 games all perform within 0.001. BPV uses decay 0.5 and k = 123. Blending in 123 games (about a season and a half) means a player's own record needs more than that before it outweighs the league average. The formula also uses real schedules (66, 64–75 and 72 games in 2011-12, 2019-20 and 2020-21), handles trades, and counts a whole missed season (like a torn Achilles) as fully missed.

Verdict: Weighted-and-shrunk history is the best method, but **availability is mostly unpredictable**: this season's and next season's availability correlate at only 0.24, and the best method beats simply assuming every player is average by about one percentage point of games. Career GP% adds nothing over the average.

8. Injury risk and the Cox model

Aryan Jha's article "Figuring out why NBA players get injured using Cox Proportional Hazards" (Medium, September 28, 2025) studied every All-Star since 2005-06: 120 players, 1,420 player-seasons and 70 season-ending injuries. Significant factors were height, season, field goals made, rebounds, blocks, fouls committed, fouls drawn and points. Each extra inch of height carried about 31% more hazard; a 6'3" player had about a 98.7% chance of finishing 82 games without a season-ending injury vs 89.4% for a 6'11" player.

$$h(t \mid x) = h_0(t) \times e^{\beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p}$$

- **How it works:** the model fits one weight β per trait across all players at once. It does not match a player to similar historical players.
- **Why it can't be multiplied into BPV:** $e^{\beta x}$ is a relative risk; above 1 means riskier. Multiplying FPTS by it would raise risky players' values.
- **Correct use:** convert risk to a probability of staying healthy, $S(t \mid x) = S_0(t)$ raised to the power $e^{\beta x}$, and use it as availability p. Or rank players by risk into percentile buckets and give each bucket its historically observed games-played share. Multiplying by the raw percentile doesn't work: it would give the riskiest player about ×0.05 and ignores how large the gaps really are.
- **Limits:** it models only season-ending injuries in All-Stars, it uses minutes and points as inputs (partly penalizing production), and the article text reports no out-of-sample accuracy.

We built and tested it

Injury reports aren't needed to try it: box scores record absences, which is what costs a fantasy manager. The model predicts when a player's first long absence of the season starts (10+ straight missed team games, any cause) from height, weight, age, last season's minutes, usage, games played and production, BPV's availability estimate, and his history of long absences. Each season 2022-23 to 2025-26 was predicted with a model fitted only on earlier seasons; risk was converted to availability through percentile buckets.

Measure	Cox model	BPV's current estimate
Ranking who misses time (C-index, 0.5 = random)	0.53–0.60	0.56–0.61
Availability error	0.170	0.166
Cox added on top of current	0.166 (difference +0.0002, interval -0.0006 to +0.0009)	—

The same happened with long absences defined as 5+ or 20+ games. Height was the only factor with a hint of signal (11% more risk per standard deviation, p = 0.08), in the same direction as Jha's finding.

Verdict: The Cox model **doesn't beat** BPV's current availability estimate, alone or combined, so it isn't used.

Official injury reports

We added the NBA's official injury reports: one per game day, 2021-22 to 2025-26 (816 days, about 60,000 listings), read with the nbainjuries package. Unlike box scores, they give a status and a reason.

Finding	Result
Why regulars (20+ mpg) miss games	injury 81%, rest 6%, G League 4.5%, COVID protocols 3.6%, personal 2%
Rest over time	2.6% of absences in 2021-22 → 9.3% in 2025-26, despite the 2023 Player Participation Policy
Share who played, by status	Probable 91%, Questionable 49%, Doubtful 1.7%, Out 0.2%

Finding	Result
Out with a serious injury	played a median 7% of remaining games; 61% missed 80%+
Their own availability	error 0.60 with BPV's usual estimate → 0.16 using the report (gain interval 0.18–0.60)
Teammate boost from those listings (no hindsight)	-0.25 for affected players (-0.40 to -0.11)
Any "Out" listing as the trigger	-0.02 (tie)
Injury and rest history → next season's availability	-0.0004 (tie)
Resting as a habit	year-to-year correlation 0.20; under 1 rest game per regular per season

Verdict: In season, BPV now **reads the latest official injury report automatically**: a player Out with a serious injury is treated as missing about 93% of remaining games, and his 20+ minute teammates get +0.029 FPTS/game per lost minute. Statuses and chances of playing are shown on the site.

9. How we tested

Every percentage in BPV was chosen by **backtesting**: pretending to stand at a point in a past season, making a prediction with only the information available then, and checking it against what actually happened.

Data

- Every regular-season player-game and team-game from 2019-20 through 2025-26 (about 26,000 player-games per season), from stats.nba.com via nba_api.
- Every NBA draft pick, for rookie baselines.
- NBA Cup championship games don't count as regular-season games and are excluded; earlier Cup games are included.

In-season tests (recent form, New Role Value, early season)

- **Checkpoints:** every 7 days, from 4 weeks into each season until 3 weeks before the end.
- **At each checkpoint,** for each player with at least 10 games before and 10 after, and at least 12 minutes per game: season-to-date average, last-5 average, minutes, and the **answer**: his actual average over the rest of the season.
- **Error** = average absolute difference between the prediction and the answer, in FPTS per game.
- **Tuning seasons** 2021-22 to 2024-25 (21,427 player-checkpoints) choose the percentages. The **test season** 2025-26 (5,404 checkpoints) is never used for tuning; results there show whether a choice really works.

Resampling (how sure are we?)

A single number like "15% is best" can be luck. To measure uncertainty we used the **bootstrap**: redraw the players at random (with replacement), keeping all of each player's checkpoints together, and redo the calculation 300 times. The spread of the answers gives a **95% interval**. If the interval for a difference includes zero, the two options are statistically tied.

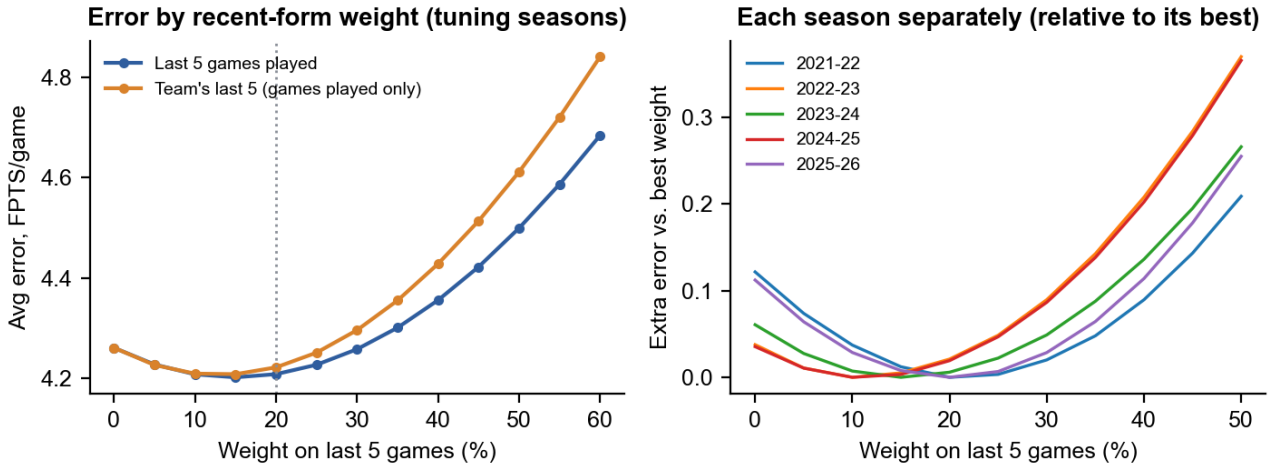
Preseason tests (model comparison)

- Each model ranks players using only seasons before the target season. Its settings are tuned only on even earlier seasons, so no model sees the answers.
- **Rank correlation:** how well each model's order matches the actual order of total FPTS (1 = perfect).

- **Simulated drafts:** 1,000 drafts per season, 10 teams, 13 rounds, snake order. Each model gets 2 randomly placed seats and takes the best player left on its list. Teams are scored by their players' actual season FPTS.
- Players who never played again after the predicted season (mostly retirements) are left out, since a preseason ranker would know about retirements.
- **Preseason production** (Section 14): 1,151 player-seasons, each target season predicted with settings fitted only on the other three (leave-one-season-out).

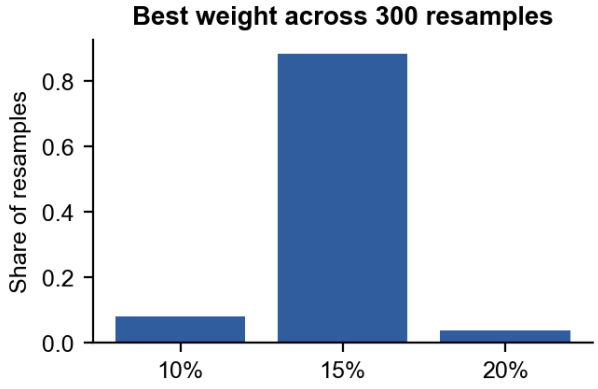
10. Recent form: finding the ideal percentage

The proposal: **BPV = 80% season performance + 20% last 5 games**. We tested every weight from 0% to 60% in steps of 5%.



Left: error across the tuning seasons for two definitions of the last 5 games. Right: each season separately.

Weight on last 5	0%	10%	15%	20%	30%	50%
Avg error (FPTS/game)	4.261	4.208	4.202	4.208	4.258	4.499



Across 300 resamples of players, the best weight was 15% 88% of the time.

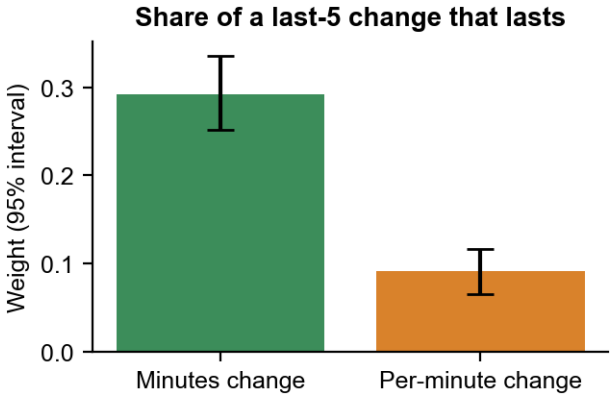
- **Recent form helps:** on the unseen 2025-26 season, 20% beat 0% by 0.11 FPTS per game (95% interval 0.05–0.17), about 3% less error.
- **The exact weight is flat:** the best weight in each season was 2021-22 20%, 2022-23 10%, 2023-24 15%, 2024-25 10%, 2025-26 20%.
- **20% vs 15% on the unseen season:** difference -0.008 (interval -0.024 to +0.005), a tie.

- **Next two weeks only** (start/sit decisions): the best weight was 20–25%, since short-term predictions lean more on recent games.
- **Which 5 games:** the player's last 5 games played beat "the team's last 5 games, counting only the ones he played" (the first draft's definition), and avoids basing recent form on 1–2 games. Absences are handled by availability instead.
- **Offseason:** in preseason tests, the 80/20 blend predicted the next season no better than the plain season average, because the end of a season is distorted by resting and tanking. BPV turns recent form off between seasons.
- **Exponential decay** (every game counting a little less than the next, as DARKO and EPM do) tied the current method on the unseen season (4.10 for a 20-game half-life vs 4.10 for BPV). The best version updated minutes fast (12-game half-life) and per-minute production slowly (30), which independently supports the minutes/shooting split in Section 11. Its gain over BPV was not significant (−0.013, interval −0.046 to +0.020).

Verdict: Giving the last 5 games played **10–20%** weight is best; the data cannot choose between values in that range. 20% is a sound choice. Recent form counts at about 28% by midseason, because the last 5 games are also part of the season average.

11. Minutes vs shooting: which streaks last

Any last-5 change can be split into a **minutes part** (per-minute production × change in minutes) and a **per-minute part** (everything else: shooting, efficiency, luck). We estimated how much of each part carries into the rest of the season.



Weights with 95% intervals. The intervals don't overlap.

- **Minutes:** 0.29 of the change lasts (interval 0.25–0.34).
- **Per-minute:** 0.09 lasts (interval 0.07–0.12).
- This matches outside research: minutes are mostly a coach's decision and stabilize within about 1.5 games, while three-point percentage takes about 240 attempts to stabilize (Medvedovsky, 2020). Research on the hot hand finds a real but small effect (about 1–2 percentage points on the next shot).
- As a prediction method, the split beat plain 80/20 on the unseen season by only 0.02 FPTS per game (interval -0.044 to +0.002), which is not significant.

Verdict: Minutes changes last about three times more than shooting swings: **established**. This is the basis for buy-low / sell-high decisions and the Waiver Wires page. BPV uses the split (0.3 minutes, 0.1 per-minute) for production; plain 80/20 would be nearly as accurate.

Buy low, sell high

Sell a player who is hot **in the same minutes** (his streak will likely fade) and buy one who is cold in the same minutes (he'll likely recover). A player whose **minutes jumped** is a real pickup. The reason to accept short-term losses for long-term value is the fantasy playoffs at the end of the season, where the championship is decided.

Example. Player L had a hot week but has never been that good; player M, a 2nd-round pick, played like a 5th-rounder for a week. BPV valued L at 23.0 and M at 25.2, and the trade for M was accepted. The direction fits the evidence (M's longer record should count more), but the 2.2-point gap is within BPV's typical error of about 4 points per game, so the example supports the method's logic rather than proving it.

12. New roles: testing the 90% and 50/50

Player N plays 10 minutes a game for 50 games, then a starter gets hurt and N plays 30. His first 50 games understate his new value. Proposal:

New Role Value = FPTS per minute × new minutes per game × haircut

BPV = (1 − w) × previous value + w × New Role Value

We tested every haircut from 0.60 to 1.10 and every weight w from 0% to 100% on the 703 cases where a player's last-5 minutes were at least 8 above his season average (from 24 minutes or less). These averaged 17.5 minutes before and 28.1 in their last 5 games.



Star: best fit (0.90 haircut, 60% weight). Circles: every pair within 1% of the best.

Method (unseen 2025-26 season)	Avg error (FPTS/game)
Season average only	7.85
80/20 recent form	6.36
New Role Value, 0.90 haircut, 50/50 (first draft)	5.36
New Role Value, 0.90 haircut, 60/40 (best fit)	5.24

- New Role Value beat 80/20 by 1.0 FPTS per game on the unseen season (95% interval 0.5–1.4), and cut error by a third compared with the season average.
- The haircut and the weight trade off: a smaller haircut with more weight fits about as well as a bigger haircut with less weight. Resampling players, the best haircut ranged over 0.80–1.10 and was 0.90 only 23% of the

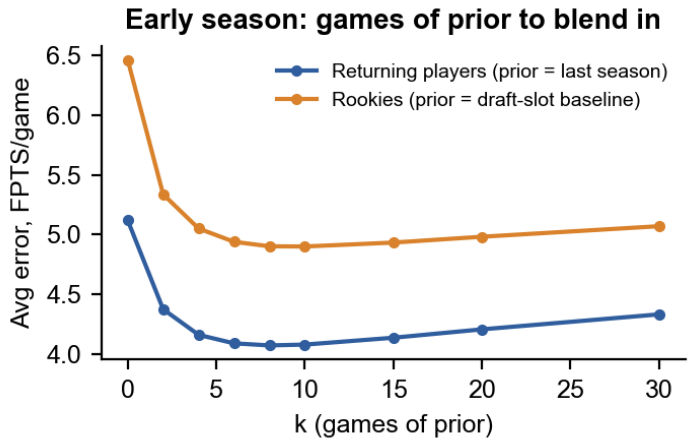
time.

- **Detection:** the site applies New Role Value automatically when a player has 10+ games, averaged 24 minutes or less, and his last-5 minutes are 8+ above his average (these thresholds define the test group; they weren't tuned). A role change can also be entered manually with a start date, since an algorithm can't tell a one-week absence from a season-long one.

Verdict: New Role Value is **clearly better** than 80/20 after a minutes jump: established. The exact haircut is **not established**; anything from about 0.80 to 1.10 with a matching weight works. BPV uses the best fit, 0.90 with 60/40.

13. Early season, rookies, and players with no games

Five games tell you little. We tested blending a player's current average with k games' worth of a prior: last season's average for returning players, and the average rookie from his draft range for rookies.



Error at checkpoints with 20 or fewer games played, by games of prior blended in.

Group	No blending (k = 0)	Best k	Error at best k	Observations
Returning players	5.12	8 games	4.07	14,278
Rookies	6.46	10 games (8 ties)	4.90	3,312

Rookie baselines (average rookie season by draft range, drafts 2020–2025):

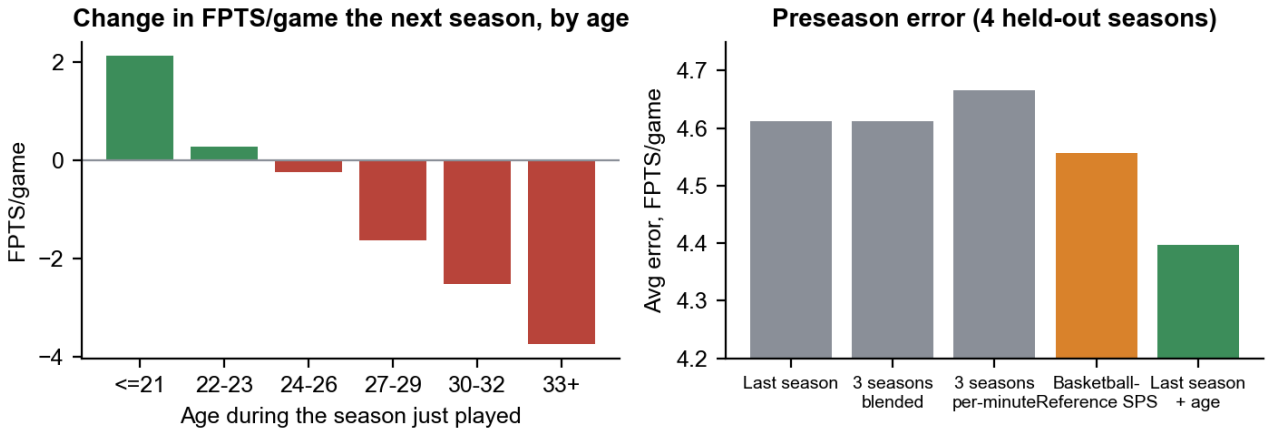
Pick range	Rookies	FPTS/game	Minutes/game	FPTS per minute	Avg games played
1-3	21	31.0	29.3	1.05	61
4-10	46	18.2	21.4	0.85	58
11-20	65	15.6	18.3	0.86	51
21-30	65	12.0	15.1	0.81	44
31-60	195	8.8	11.2	0.80	26
Undrafted	287	8.2	11.0	0.81	14

- By about game 24, a returning player's current season carries 75% of the weight (24 of 32 games).
- Players with no NBA games yet (new draftees) get their draft range's baseline, league-average availability and a "projection only" label. The first version of the site gave them 0.

Verdict: Blending cuts early-season error by about 20–25%: **established**. BPV uses 8 games of last season for returning players and 10 games of the draft baseline for rookies.

14. Between seasons: age matters, older seasons don't

Before a season, BPV projects each player's production from what he did before. We compared six ways of doing that on 1,151 player-seasons (20+ games in both seasons, 15+ minutes per game), predicting each season from 2022-23 to 2025-26 with settings fitted only on the other three.



Left: the age curve BPV uses. Right: preseason error of each approach.

Preseason production model	Error (FPTS/game)	Result
Last season's FPTS per game (BPV before this test)	4.61	Baseline
Last three seasons blended, shrunk toward the average	4.61	Tuning chose last season only
Three seasons of per-minute production × last season's minutes	4.67	Significantly worse
Basketball-Reference's Simple Projection System	4.56	Slightly better than baseline (not significant)
Last season + age adjustment	**4.40**	**Best**

- **Age:** cut error by 0.21 FPTS per game (95% interval 0.12–0.31), about 4.7%, and helped in all four seasons.
- **Benchmark:** it beat Basketball-Reference's projection system by 0.16 FPTS per game (interval 0.04–0.27).
- **Flat vs proportional:** adding a fixed number of points by age beat scaling by a percentage (4.40 vs 4.43), and did better for stars.
- **Stability:** the curve fitted only on seasons before 2022-23 is nearly the same as the final one.
- **Older seasons don't help,** tested two ways. This goes against standard practice (Basketball-Reference and baseball's Marcel blend three seasons). A likely reason: a fantasy player's value is mostly his current role, and older seasons describe roles he no longer has.
- **Caveat:** players who fell out of the league were excluded, which probably understates the decline at older ages.
- **Early season too:** age-adjusting last season's level in the early-season blend (Section 13) cut error by 0.05 FPTS per game (interval 0.02–0.08).

Roster competition

Next season's rosters are public before the season. For each player we measured **crowding**: the total last-season minutes per game of his new teammates who played 10+ minutes, minus the league-average team.

- Crowding cut preseason error by a further 0.06 FPTS per game (interval 0.02–0.10), with the same coefficient in all four held-out seasons: **-0.0177 FPTS per game per minute**. Joining a team with 40 more rotation minutes than average costs about 0.7 FPTS per game.
- Just knowing he changed teams didn't help (-0.02, interval -0.04 to +0.00); the competition for minutes is what matters.
- Preseason rosters carry about 20 players per team vs about 15 on opening night, but only 10+ minute players count, so the scale matches the test closely.

Verdict: Between seasons, BPV uses **last season + a flat age adjustment + roster crowding** (for players with 20+ games and 15+ minutes). Established: better in every season, and better than a published projection system.

Head to head with ESPN

ESPN's public fantasy API still serves its preseason projections for past seasons (2021-22 and 2023-24 to 2025-26; its 2022-23 file was frozen mid-season and is excluded). This is the competitor comparison that was missing. Everyone who qualified the season before is included, scored on actual season totals (0 if he then missed the season).

Preseason projection	Error on season totals	Ranking accuracy	Draft sims (avg team FPTS)
ESPN	576	0.671	27,466 (won 1 of 4)
BPV alone	502	0.664	26,507 (won 0)
BPV + ESPN hybrid	496	**0.686**	**28,523 (won 3 of 4)**

- Per game, BPV beats ESPN (4.36 vs 4.55 error; ESPN over-projects by about 1.6 FPTS/game), but a 62.5/37.5 blend beats both.
- ESPN's projected games carry preseason injury, suspension and holdout news that BPV can't see, so ESPN out-ranked plain BPV on season totals. But it projects too many games (median 66–72), which hurts its totals.
- The hybrid uses the per-game blend with ESPN's projected games scaled by 0.86: it ties BPV on total error, beats ESPN by 80, ranks best, and won 3 of 4 draft seasons.

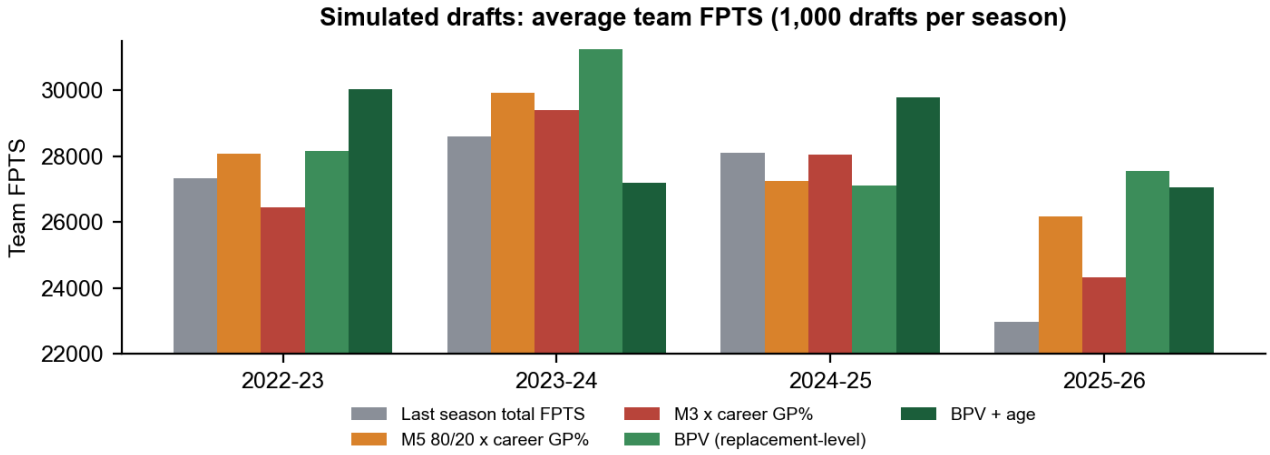
Verdict: Plain BPV was **not** the best preseason ranking; ESPN was. BPV now uses the **BPV + ESPN hybrid** before each season (veterans with 20+ games and 15+ minutes). It isn't used in season, where ESPN switches to rest-of-season projections.

15. Which preseason model is best?

Each model below predicts a season using only earlier seasons, for four seasons (2022-23 to 2025-26).

Model	Rank correlation (avg)	Draft sim: avg team FPTS	Draft-sim wins
Last season's total FPTS (like a naive consensus)	0.638	26,744	0 of 4
Model 1: season FPTS per game	0.644	not simulated	—
Model 3: × career GP% (first version of BPV)	0.651	27,050	0 of 4
Model 5: 80/20 × career GP%	0.654	27,851	0 of 4
Revised BPV: p × F + (1 − p) × q × R	0.651	28,514	2 of 4
Revised BPV + age (Section 14)	0.665	28,515	2 of 4

BPV here uses the league-average q, as the site does, and each season's age curve was fitted only on earlier seasons. Roster crowding was added after these simulations; its accuracy gain was tested separately (Section 14).



Average team FPTS per model in 1,000 simulated drafts per season.

- **Ranking:** BPV + age has the best average rank correlation. It beats BPV without age significantly in 2 of 4 seasons (tied in the rest), but beats Model 5 significantly in only 1 of 4.
- **Drafts:** a BPV version (with or without age) came first in all four seasons, and the two versions split the wins. The age curve added nothing in drafts.
- **Limits:** four seasons is a small sample, and the simulations shuffle draft order but can't replay how players actually performed.
- A "managed" score that credited free-agent points for long absences was dropped from the evidence because it builds in BPV's own assumption.

Verdict: BPV + age is **at least as good** as every alternative and best on average, and a BPV version won every season of simulated drafts. Evidence that it is **better** is growing but not conclusive.

The original preseason experiments

Before 2025-26, five sources were compared. **Experiment 1** (each drafts its ideal 13 alone): ESPN 43,565; BPV 43,113; professionals 41,851; my rankings 41,119; ADP 40,904. **Experiment 2** (all five in one draft, random order): BPV 39,096; my rankings 38,058; ESPN 37,420; ADP 32,752; professionals 27,891.

- Experiment 1 is essentially a tie (a 1% gap). Experiment 2 is a single draft, where draft position alone can swing results by thousands of points.
- Even a nearly perfect list doesn't guarantee the best team, because you don't know who will be left at your pick.
- If the preseason BPV came from the first version of the site, it was season FPTS × career GP% plus a hand-made list of 40 players set to 0, and its "last 5 games" came from the end of 2024-25.
- Summing 13 players' totals ignores that only 10 start each day and that leagues are decided by head-to-head playoffs.
- To redo the experiments properly: save each ranking (ESPN, ADP, experts, personal) as a CSV of names and run **python -m analysis.draft_sim --ranking ESPN=espn.csv ...**, which runs 1,000 simulated drafts.

16. Every improvement we tested

The rule was set before testing: a change is added only if it beats current BPV on seasons it wasn't fitted on, with a 95% interval (resampling whole players) that excludes zero. Display features are added only if they are correct and harmless.

Candidate	Result vs current BPV (FPTS/game)	Decision
Age adjustment between seasons	−0.21 (−0.31 to −0.12)	**Added**
Roster crowding between seasons	−0.06 (−0.10 to −0.02)	**Added**
Age-adjusted early-season blend	−0.05 (−0.08 to −0.02)	**Added**
Known long-term teammate absence (in season)	−0.52 for affected players (−0.79 to −0.24)	**Added** (manual injuries)
Same, guessed from missed games	+0.00 (−0.03 to +0.03)	Not added
Teammate out for one game	−0.10 per game (−0.14 to −0.06)	Not added: needs a live injury feed
Cox injury model	+0.004 alone; +0.0002 combined	Not added
Averaging with Basketball-Reference's projections	−0.03 (−0.06 to +0.01)	Not added (tie)
Blending older seasons (two ways)	+0.00 and +0.05	Not added
Exponential decay instead of last 5	−0.01 (−0.05 to +0.02)	Not added (tie)
Last 3, 7, 10 or 15 games	all within ±0.02	Not added (tie)
Separate recency per stat	+0.03 (+0.00 to +0.05)	Not added (worse)
Separate age curve per stat	identical, or +0.03 if proportional	Not added
Changed-team flag	−0.02 (−0.04 to +0.00)	Not added (tie)
Positional replacement levels	gaps under 0.3; drafts +1.4% (−3% to +4%)	Not added
80% likely range	79.8% coverage in season; 72–82% between seasons	**Added** (display)
Fantasy-playoff game counts	−0.8% error on playoff totals (−1.7% to −0.04%)	**Added** (display)
League and roster size settings	sets the replacement level	**Added**
Serious injury on the report → long absence	availability error 0.60 → 0.16	**Added** (automatic)
Teammate boost from serious-injury listings	−0.25 for affected (−0.40 to −0.11)	**Added** (automatic)
Teammate boost from any Out listing	−0.02 (−0.11 to +0.06)	Not added (tie)
Injury/rest history → next season availability	−0.0004 (−0.0014 to +0.0005)	Not added (tie)
Status → chance of playing	60,000 listings	**Added** (display)
In-season recalibration	−0.07 pooled (−0.12 to −0.02); 3 of 5 seasons better, none worse	**Added**
In-season age drift	−0.06 (−0.16 to +0.04)	Not added (tie)
Per-stat stabilization padding	-0.01 (-0.08 to +0.06)	Not added (tie)
Lighter shrinkage for stars	+0.00 (−0.01 to +0.01)	Not added (tie)
Late-season tanking/resting	-2.8 FPTS per playoff total (all 5 seasons better)	**Added** (expected playoff FPTS)
Rookie draft age	−0.02 (−0.07 to +0.04)	Not added (tie)
Usage-weighted teammate boost	+0.08 (−0.00 to +0.17)	Not added (tie)

Candidate	Result vs current BPV (FPTS/game)	Decision
Contract-year effects	not tested: no free contract data	—
BPV + ESPN hybrid (before the season)	ranking +0.016 vs ESPN; totals −80 vs ESPN, tie vs BPV; drafts 3 of 4	**Added**
Preseason minutes, veterans	-0.02 (-0.06 to +0.02)	Not added (tie)
Preseason minutes, rookies	-0.21 (-0.40 to -0.03)	**Added**
Return timelines by injury type	−0.02 absolute, −0.04 squared error	**Added** (means)
Team minutes scaled proportionally	+0.015 (tie)	Not added
Weekly totals with the injury report	-2.3 per week (-2.8 to -1.8)	**Added** (this week)
One-IR-slot draft simulation	current streaming credit drafted best	Formula confirmed
Coaching changes	−0.005 (−0.016 to +0.009)	Not added (tie)
Weekly consistency	weak trait; <1 point of H2H win odds	Not added
Weekly: team-aware report lookup + sat recently	−2.2/week (−2.5 to −2.0), 5 of 5 seasons	**Added**
In-season shooting regression (own level)	−0.064 (−0.096 to −0.036), 4 of 5; better for stars	**Added**
Shooting regression toward league average	−0.048 but hurt elite players (4.12 → 4.65)	Rejected
Shooting regression between seasons; FT% regression	ties	Not added
This week's opponents	−0.04/week, 5 of 5 (negligible)	**Added**
Tracking: deflections, potential assists, rebound chances, contests	stat-level gains; FPTS tie	Not added
Shot quality by defender distance	nothing beyond plain eFG%	Not added
Stat-specific crowding; scorekeeper generosity; forecast disagreement	ties / negligible	Not added
Post-injury ramp-up; participation policy; back-to-backs; schedule strength	ties	Not added
Returning teammate; mid-season trades; second-year jump; pick development; second-half weighting; 65-game rule	ties	Not added
Preseason projection (BPV + ESPN) as early-season prior	−0.064 (−0.102 to −0.025), 3 of 4 seasons	**Added**
Rookie preseason minutes in-season; smooth age curve	ties	Not added
Replacement level 100th/160th/190th/220th player	all below 130th in standard draft sims	Not changed
In season: current absence streak	−0.0034 (−0.0049 to −0.0023); 4 of 5, squared 5 of 5	**Added**
Next-season availability: 23 injury-proneness signs (age, minutes, playoff minutes, past injuries, body parts, size, usage, ...)	all ties or worse, vs history and vs ESPN	Not added
In season: this season 2–5×, age, report status, non-injury Out, recent return	ties or worse	Not added

Holdout test (2014-15 to 2018-19, never used before): Today's formula, frozen, with pass criteria written first. 8 of 9 checks passed: preseason ranking beat last season's totals (4 of 5 seasons); age + crowding helped in 5 of 5 (per-game error -0.19); in-season BPV beat the original 80/20 in 5 of 5 (3.64 vs 3.84 FPTS/game); recalibration 4 of 5, shooting regression 5 of 5, early prior 5 of 5, availability 5 of 5, absence streak 5 of 5; drafts beat naive 4 of 5 and the original 3 of 5. Failed: preseason ranking vs the original formula, better in only 3 of 5 (0.708 vs 0.697). The ESPN blend and injury-report features couldn't be tested on those years.

Stopping point: The loop stopped when the last round added nothing and the remaining ideas couldn't be tested: xFG% and gravity exist only from 2025-26; referees, travel, altitude and play-type matchups are single-game effects; betting props and ESPN in-season projections have no free history; college, G League and contract data have no free source. About 105 ideas were tested on the same five seasons, so some small wins may be luck; 2026-27 is the real test.

Candidates came from our own ideas and two rounds of research into published systems (DARKO, EPM, FiveThirtyEight, Basketball-Reference, SCHOENE) and peer-reviewed work. Public out-of-sample evidence on season-long fantasy projections is scarce. Notable: BPV was slightly too extreme in season (players under 24 improve during a season by 1.9–2.7 FPTS/game); on bad teams, young regulars play 24 points less of their games in the fantasy playoffs and veterans 44 points less.

17. The final model: every setting and where it came from

Setting	Value	How it was chosen	Confidence
Scoring weights	ESPN default	ESPN's published scoring	Fact
Recent window	Last 5 games played	Beat the team-window version; 3, 7, 10, 15 games tied	Strong (flat)
Recent weight	0.3 minutes / 0.1 per-minute (≈ 20% overall)	Fitted on 2021-22 to 2024-25	Range 10–20% supported
Recent form in offseason	Off	Preseason tests: no benefit	Moderate
New role detection	10+ games, ≤ 24 min, last-5 minutes +8	Definition (not tuned)	Judgment call
New role haircut / weight	0.90 / 60%	Best fit on 546 cases	Haircut anywhere 0.80–1.10
Early-season prior	8 games of last season	Best of k = 0–30	Strong
Between-season production	Last season + flat age adjustment (+2.1 at ≤21 to −3.7 at 33+)	Leave-one-season-out, 6 approaches compared	Strong
Roster crowding	−0.0177 FPTS/g per minute of competition	Leave-one-season-out	Strong (same in all 4 seasons)
Injured-teammate boost	+0.034 FPTS/g per lost minute (in season, known injuries)	Fitted 2021-22 to 2024-25, tested 2025-26	Strong for known injuries
Early-season prior age	Last season + age adjustment	11,587 checkpoints	Strong
80% range	In season −5.5 to −5.8 / +7.2 to +8.2 by games; between seasons −7.3 / +6.2	Out-of-sample residuals	Calibrated
In-season recalibration	2.96 + 0.91 × projection (10+ games)	Leave-one-season-out	Moderate (3 of 5 seasons)
Early-season prior	Saved preseason BPV + ESPN projection, 8 games' weight	ESPN seasons 2021-22..2025-26	Moderate (3 of 4 seasons)
Absence streak (in season)	+0.0075 − 0.0242 × team games missed in a row (max 20), 20+ mpg, not listed Out injured, 10+ games left	10,139 checkpoints, leave-one-season-out	Strong (4 of 5; squared 5 of 5)
In-season shooting regression	−1.85 × (eFG% − his last-season eFG%) × FGA/g	All weekly checkpoints, leave-one-season-out	Strong (4 of 5; stars improved)
Weekly status lookup	Latest report with his team's game + sat-in-last-3-days factor	Leave-one-season-out	Strong (5 of 5)
Serious injury on report	Teammates +0.029/min (own availability: see injury-type timelines)	Fitted 2021-22 to 2024-25, tested 2025-26	Strong
Fantasy-playoff games played	Share change by team status and age (bad veterans −44 points)	Leave-one-season-out	Strong (5 of 5 seasons)
ESPN blend (before the season)	37.5% of per-game; games = ESPN × 0.86	Leave-one-season-out, 4 seasons	Strong on ranking

Setting	Value	How it was chosen	Confidence
Injury-type timelines	Share of remaining played: serious 24%, sprain 53%, strain 58%, soreness 54%, other 40%	5 seasons of reports	Strong
This week's value	Production × games × share × status factor (e.g. Out-serious 0.09, Questionable 0.94)	Fitted 2021-22 to 2024-25, tested 2025-26	Strong
Rookie preseason minutes	+0.31 FPTS/g per preseason minute above his draft range's typical	Leave-one-season-out	Moderate
Rookie prior	10 games of draft baseline	Best of k = 0–30 (8 ties)	Strong
Availability history	Each older season × 0.5	Best of 5 options	Flat: 0.5–0.7 equivalent
Availability shrinkage	123 games of league average	Best of k = 0–246	Flat: 123–164 equivalent
Long absence	5+ straight games	Judgment (3 and 10 also tested)	Choice
q (IR-able share)	League average ≈ 72%	Measured	Per-player values unreliable
Replacement rank	Teams × roster size (130 by default)	League settings	Depends on league size
History used	Last 5 seasons (availability); last season (production)	Older seasons tested: no gain for production	Moderate
Heat Check Index	BPV ÷ best BPV × 100	Display measure	—

18. Limitations and what to try next

Limitations

- Box scores can't tell injuries from rest, suspensions or coach's decisions, which limits every availability measure.
- Availability is mostly unpredictable (year-to-year correlation 0.24); BPV can only price it on average.
- BPV's typical error for one player's rest-of-season average is about 4 FPTS per game, so close comparisons are uncertain.
- Positions are ignored (tested: positional replacement levels differ by under 0.3 FPTS/game with guard/forward/center data).
- Injury reports exist only on game days and don't give return dates; long absences are inferred from the injury type.
- stats.nba.com blocks many cloud hosts, so the site runs locally (hosting would need a proxy or a pre-filled cache).

What to try next

- **1. Score the BPV + ESPN hybrid against competitors in 2026-27.** About 105 ideas were tested on the same five seasons, so a fresh season is the real test. Run **python -m analysis.snapshot** before October 20 and save ESPN's and other rankings the same week; score them all after the season with analysis/draft_sim.py. A peer-reviewed study found professional fantasy projections beat naive ones only moderately, so expect small differences between good systems.
- **2. A full minutes model** (depth charts, as FiveThirtyEight built). Crowding is a first step; projecting each player's minutes directly could capture role changes without roster changes.
- **3. A fuller weekly view** adding the game-level teammate effect (tested at game level, not yet built).

- **4. Finer injury types** (e.g. hamstring vs calf) once more seasons of reports accumulate.
- **5. Contract-year effects**, if clean contract data can be found (published evidence is mixed).
- **6. Rest-of-season trade values** weighted by each team's remaining schedule.
- Everything else on the earlier list was tested (Section 16).

Appendix A: What changed from the first version

First version	Now	Why
BPV = FPTS × career GP%	BPV = p × F + (1 − p) × q × R	Missed games aren't worth zero
Career GP%, assuming 82 games	Weighted, shrunk, real schedules	Career GP% wasn't better than the average; shortened seasons and trades were miscounted
Reliability score undefined	League-average q + manual injuries	Per-player absence patterns can't be measured reliably
Cox × FPTS	Cox → availability (future)	Hazard ratio above 1 means riskier
Team's last 5 games	Last 5 games played	Tested slightly better
20% (informal)	Tested: 10–20% supported; minutes/per-minute split	Measured, with intervals
NRV 90%, 50/50, manual	0.90, 60/40, automatic + manual	Idea confirmed; haircut range 0.80–1.10
Rookies = 0	Draft-slot baselines	Players without history were ignored
40 players hard-coded to 0	Manual overrides file	The list was stale and name-matching failed
One draft per experiment	1,000 simulated drafts × 4 seasons, no peeking	One trial can't separate skill from luck
Last season, no age	Last season + age adjustment between seasons	4.7% less preseason error; beat Basketball-Reference's system
Last season's team shown	Next season's rosters + roster crowding	Offseason moves were invisible; 1.4% less error
Cox proposed	Cox built and tested	Didn't beat the current availability estimate
No uncertainty shown	80% likely range + fantasy-playoff games	Calibrated; playoff counts matter for head-to-head
Injuries entered by hand only	Official injury report read automatically in season	Serious injuries predict long absences with no hindsight
No competitor comparison	Compared with ESPN; BPV + ESPN hybrid before the season	ESPN out-ranked plain BPV; the hybrid beat both
Client-side math, two conflicting copies	One server-side calculation	Pages disagreed; the 80/20 blend never actually ran
nba_api 1.7.0	nba_api 1.11.4	The NBA site had started rejecting the old version, so no data loaded

Appendix B: Reproducing everything

- **python -m analysis.backtest**: all tuning results (analysis/results.json).
- **python -m analysis.robustness**: intervals, per-season results, no-peeking comparison, absence-pattern reliability (analysis/robustness.json).
- **python -m analysis.draft_sim**: simulated drafts, optionally with your own ranking CSVs.
- **python -m analysis.preseason_production**: the between-season comparison and age curve (analysis/preseason_production.json).
- **python -m analysis.decay_test**: exponential decay vs the last-5 rule.
- **python -m analysis.snapshot**: saves today's rankings to data/snapshots/ for scoring after the season.
- **python -m analysis.test_espn_projections**, **test_espn_totals**, **test_round3**, **test_coach_consistency**, **test_injury_reports**, **test_research_ideas**, **test_research_followup**, **test_cox**, **test_minutes_model**,

test_misc, **test_teammates_perstat**, **test_schedule_positions_uncertainty**: the Section 16 tests.

- **python -m analysis.report_charts DIR** and **python -m analysis.report_pdf DIR**: this report.
- Shrinkage formula used throughout: estimate = (n × own + k × prior) ÷ (n + k), standard empirical-Bayes weighting.

References

- ESPN. "ESPN Fantasy's default points-league scoring explained" (2025); "ESPN Fantasy Basketball 101: Adjusting the settings"; ESPN Fan Support articles on IR eligibility and waiver order.
- Jha, A. (2025, September 28). "Figuring out why NBA players get injured using Cox Proportional Hazards." Medium.
- Herzog, M. M., Brink, A., Gondalia, R., DiFiori, J. P., & Mack, C. D. (2026). The relationship between games missed for rest or load management and injury in the NBA, 2014–15 through 2022–23. Sports Medicine 56(9), 2313–2323.
- Menon, A. et al. (2024). Season-ending injuries in the NBA. Arthroscopy 40(9), 2468–2473.
- Medvedovsky, K. (2020, August 6). "NBA stabilization rates and the padding approach." kmedved.com.
- Miller, J. & Sanjurjo, A. (2018). Surprised by the hot hand fallacy? A truth in the law of small numbers. Econometrica 86(6), 2019–2047.
- Bocskocsky, A., Ezekowitz, J. & Stein, C. (2014). The hot hand: a new approach to an old "fallacy." MIT Sloan Sports Analytics Conference.
- Efron, B. & Morris, C. (1975). Data analysis using Stein's estimator and its generalizations. JASA 70(350), 311–319.
- Davidson-Pilon, C. lifelines documentation (CoxPHFitter; proportional hazards checks).
- Bryant, J. "Principles of Value Based Drafting." Footballguys.
- Basketball-Reference. "Simple Projection System." basketball-reference.com/about/projections.html.
- Döpke, J., Köhler, T. & Tegtmeier, L. (2024). Forecasting performance of NBA daily fantasy projections. Journal of Economics and Finance 48(1).
- Medvedovsky, K. DARKO documentation, darko.app/about; Dunks & Threes, EPM methodology, dunksandthrees.com/about/epm.
- nbainjuries (Python package, v1.1.1); lifelines (v0.30.3); NBA official injury reports (ak-static.cms.nba.com/referee/injury).
- Pelton, K. SCHOENE projection system (ESPN); Basketball-Reference, SCHOENE vs SPS comparison (2009).
- ESPN Fantasy public API (lm-api-reads.fantasy.espn.com), preseason player projections 2021-22 to 2026-27.
- swar/nba_api (v1.11.4) and stats.nba.com.